

# Smart Irrigation System for Protected Flower Cultivation Using Real-Time Sensing and Automation

Vadlamudi Sreevanth Chowdhary  
*Department of Computer Science and Engineering*  
*Amrita School of Computing*  
Amrita Vishwa Vidyapeetham, Bengaluru, India  
vsreevanth@gmail.com

Tammalla Nikhil  
*Department of Computer Science and Engineering*  
*Amrita School of Computing*  
Amrita Vishwa Vidyapeetham, Bengaluru, India  
bl.en.u4cse22156@bl.students.amrita.edu

Yendamuri Nishanth  
*Department of Computer Science and Engineering*  
*Amrita School of Computing*  
Amrita Vishwa Vidyapeetham, Bengaluru, India  
bl.en.u4cse22164@bl.students.amrita.edu

Valupadasu Srujan  
*Department of Computer Science and Engineering*  
*Amrita School of Computing*  
Amrita Vishwa Vidyapeetham, Bengaluru, India  
bl.en.u4cse22262@bl.students.amrita.edu

**Abstract**—Water management is critical to efficient management of sustainable agriculture, particularly in the tallied flower farming where exact environmental regulated controls directly influence the quality and yield of plants. Polyhouses have traditional irrigation systems which are usually manual or timer-controlled thus resulting to wastage of water, erratic soil moisture and low crop production. To ensure that these challenges are addressed, this work suggests the Smart Irrigation System that takes advantage of real-time sensing and automation with the help of IoT and the ESP32 microcontroller. The system is based on soil moisture, temperature, humidity, pH, and turbidity sensors that are continuously used to monitor the conditions in the field. The ESP32 receives and processes the data and sends it to the Blynk IoT platform to which irrigation can be remotely controlled and controlled manually. Irrigation pump is started when it is needed and thus, the water will be used optimally and there will be the best growing conditions in the guarded building. Besides minimizing human labor, this method improves quality of crops, saves resources as well as facilitating sustainable precision farming. Its affordability, scalability, and connectivity to the cloud render the system applicable to greenhouse farming, smart farms, and the agricultural setting that is research-based.

**Index Terms**—Smart Irrigation, IoT, ESP32, Soil Moisture Sensor, Blynk App, Automated Farming, Precision Agriculture, Wireless Sensor Networks, pH Monitoring, Temperature and Humidity Control, Real-Time Data, Protected Cultivation, Sustainable Agriculture, Cloud-Based Monitoring

## I. INTRODUCTION

Most of developing economies rely on agriculture and the increasing world population requires sustainable solutions to food and floriculture production. Protected cultivation (poly houses and green houses) is one of the modern agricultural methods that are important in ensuring controlled environment in high value crops such as roses, gerberas, carnations, orchids, chrysanthemums, etc. Nevertheless, another most important

feature of the protected cultivation is the accurate management of the irrigation since slight variations of the soil moisture, temperature or pH levels can impact the quality of crops, their productivity and resistance to diseases significantly. The conventional irrigation methods that are either manually operated or programmed on a specific time schedule tend to cause wastage of water, imbalanced nutrients and irregular plant growth and development, contributing to the inefficiency of both economically and environmentally. Thus, to achieve the maximum use of resources, as well as to satisfy the growing needs of modern agriculture, introduction of smart automation and IoT technologies have become a necessity.

The Smart Irrigation System suggested in this paper will help to offer an automated and data-driven irrigation management system, viz., the one that is specific to the cultivation of protected flowers. The system constantly measures the soil moisture, temperature, humidity, pH, and water turbidity using sensors and regulate the irrigation programs based on the data obtained. ESP32 and Wi-Fi built-in microcontroller serve as the main processing and communication component that uses real-time data collected by the sensors to transmit it to the Blynk IoT platform. Relay module can automatically switch the water pump off or on to the system according to the sensor readings so that the plants do not get unnecessary water. This adaptive control avoids the over-irrigation and under-irrigation water, hence saving water and keeping the soil environment in the optimal range. Moreover, the system is remotely controlled and monitored by users via a mobile application, which increases the convenience and flexibility of the farmers.

Irrigation solution is an IoT-based solution that enhances the efficiency, which is also based on sustainable and precision agriculture through reduction of manual labor, lessening

human error, and maximising input resources. The scalability and affordability of the system enable it to be used in a broad spectrum of agricultural applications including a small nursery and a large commercial greenhouse. Moreover, it is connected to the cloud and can be easily integrated with the future technologies of machine learning-based irrigation prediction, solar-powered operation, and weather-forecast-based automation. Therefore, the Smart Irrigation System is a solution that can be proposed as the step toward intelligent, environmentally friendly, and autonomous agriculture that will transform the use of water in the controlled agricultural setting and its crop yields.

## II. LITERATURE SURVEY

Anver et al. [1] proposes an IoT-based automated irrigation system which incorporates IoT soil moisture, temperature and humidity sensors connected to an Arduino board, allowing for rapid adjustment in water supply by switching motors on when soil moisture levels dip below threshold values. The system remotely transmits soil data in real-time to an end-use device to assist farmers in monitoring field conditions away from the crop domain. This study also emphasizes both low-cost and low-energy irrigation systems for larger scale farms. Networks of sensors contribute to reliability, and are directed towards reducing water waste, for possible scaling through rural agriculture. Ambika et al. [2] focuses on irrigation automation based on IoT sensors and machine learning-driven predictive models. DHT11 and soil moisture sensors are used to evaluate temperature, humidity, and evaporation on rainy and dry days. Linear regression shows real-time watering intervals for crops effective, like roses. The system automatically controls the servo motor to open and close water valves based on real-time soil conditions. This irrigation technology is expected to save water and labor while improving crop health through predictive irrigation control.

Abhaykrishnan et al. [3] presents an IoT-based, real-time irrigation monitoring system utilizing Arduino UNO, soil moisture, and soil humidity sensors. This system implements a cloud connection into which the user can download the environmental measurements and (if they have activated) the capability to control the irrigation systems remotely. The system automates the pumps when the soil moisture has dropped below the settings the farmer or user set, contributing to long term efficiencies in water management. The system supports users at a distance from Wi-Fi networks using low-power modules. For the defined areas of environmental control, the technology provides a case for sustainable agriculture through water conservation and productivity with smart automation and management. Boddu et al. [4] comprised IoT-based irrigation management with renewable energy generation from solar and piezoelectric energy sources singularly and in combination with each other. The hybrid energy system provided power for agriculture management devices and sensors, reducing farmers' reliance on energy sources like the grid. Monitoring occurred in real-time using the ESP32 and Arduino. The cost analysis over time indicated long-term economic and environ-

mental viability over the present energy systems. The study and experimentation provide a case for integrating hybrid energy solutions as an accepted technology for sustainable agriculture, resilience, and scalability as an innovative technology of modern precision agriculture.

Abhiram et al. [5] describes an Internet of Things (IoT) platform that continuously monitors the environment using nodemcu ESP8266 microcontrollers (MCU), DHT11 sensors, and GPS modules in order to capture temperature, humidity, and location, thereby computing environmental variability in real-time. The data collected are uploaded to cloud computing platforms such as ThingSpeak for visualization and alerts. Potential applications of the proposed system could be applied in the agricultural, logistics, and environmental pollution realms. The overall design focuses on low power consumption, reliable wireless communication, and effective data analytics that will augment decision making in smart environmental applications. Archanaa et al. [6] employs Bioristor sensors, with three machine learning models to measure, assess and predict water stress in tomato plants, by capturing changes in the ionic activity of plant sap to monitor plant stress before visual symptoms appear. Subsequently, models were developed for 1-hour or 24-hour forward interval predictions using a 3-stage multi-class classification system (Healthy, Stressed, Recovery). The proposed system assists in optimizing water management strategies for irrigation in precision agriculture. Results confirm sufficient accuracy, and support, the potential for integration into automated irrigation systems.

Nur et al. [7] creates the prototype of IoT that is built on ESP32, soil moisture sensors, and Blynk Cloud to monitor real-time irrigation. It also automatically irrigates water according to the level of moisture in young rocky soil and inform the users about this through mobile applications. The system reduces the cost of water and labor and increases accuracy of irrigation. It has been tested on potted plants and is effective in that it does not need human intervention in watering. The model shows how the IoT application eases the process of irrigation and encourages resource savings. Isuri et al. [8] proposes a smart irrigation system aimed at household gardens that automates the delivery of water and fertilizer. Using soil moisture, temperature, and humidity sensors the system operates in automatic and semi-automatic modes. The user will also be able to monitor and control the system using Wi-Fi over a web interface. The system also provides optimum watering requirements for different types of plants while also minimizing the amount of human effort and water wasted. The 3D printed enclosures and simulations show that it is practical to use in a smart home gardening environment.

Samir et al. [9] outlines a wireless sensor network (WSN)-based irrigation system using ESP32 and Mini ESP32-S2 microcontrollers. It measures soil moisture, temperature, and humidity, with an ultra-low power operation. Data is transmitted wirelessly between nodes using the ESP-NOW protocol which is secure and energy-efficient. The measurement node is powered by an 8800 mAh battery, and can run autonomously for nearly two years. The WSN minimizes water wastage,

improves the accuracy of irrigation, and allows for sustainable agricultural practices in resource-limited contexts. Thushani et al. [10] presents an integration of Internet of Things (IoT) and WSN to control a drip irrigation system by using fuzzy logic control. It uses multiple sensors that measure temperature, humidity, and soil moisture to automate the control of water flow in multiple zones. The data is transmitted over the internet and to the dashboard where users are able to control the system, both manually and automatically. The system operates sustainably relying on a solar energy source. The proposed user-centric framework reduces water wastage, decreases labor costs, and improves crop yield.

Varalakshmi et al. [11] evaluated the quality of soil and water in rose growing areas of Rajasthan and Uttar Pradesh. The findings indicated that the majority of soils in Uttar Pradesh were alkaline ( $\text{pH} > 8$ ) and moderately saline ( $\text{EC} < 0.5 \text{ dS/m}$ ) and bicarbonate. The soils in Rajasthan contained high PH, chloride and sodium that could be detrimental to the growth of roses. Also noticed mild alkalinity and higher percentages of soluble sodium in water samples. The research was able to conclude that the poor quality of soil and water might hamper rose production and proposed some reclamation techniques along with salt-tolerant rootstock to conduct the practice sustainably. Monika et al. [12] studied the water quality parameters of pH (4–7) and electrical conductivity ( $\text{EC} 1\text{--}4 \text{ dS/m}$ ) of irrigation water in relation to hydrangea growth and responses to stress. Greenhouse experiment exhibited the optimal growth of the plants with pH 6 and  $\text{EC} 2 \text{ dS/m}$  and generated the highest plant biomass and the lowest stress levels (proline and MDA). On the other hand, extreme acidity (pH 4) or salinity ( $\text{EC} 4$ ) led to physiological stress and growth alleviation. The researchers found that EC is largely controlled by the pH of irrigation water and that this element has a significant role when it comes to the plant health and quality of hydrangee. The management of pH and EC can enhance aesthetics of flowers and resistance of plants in ornamental horticulture.

Rodolfo et al. [13] examined the effect of soil pH levels (5–7) on the growth and reproductive characteristics of *Ambrosia artemisiifolia* (common ragweed). Plants cultivated with acidic values of 5–6 responded to higher growth, pollen was higher and they flowered earlier than the plants at the neutral pH of 7 which neither grew nor flowered. Moreover, pH 5 plants produced more allergens indicated by an increase in the IgE-binding activity. The results indicate that not only the invasion of invasive weeds depends on the soil pH but also their allergenic ability, which has the consequence on the environment and health of people. Luis et al. [14] measured the salinity ( $\text{EC} 2\text{--}10 \text{ dS/m}$ ) and alkaline pH (6.4 vs 7.8) which effects on three marigold varieties (*Tagetes patula* and *T. erecta*). It was found that salinity of  $8 \text{ dS/m}$  had a moderate negative effect and high pH 7.8 had a dramatic negative effect of 89% on shoot dry weight in certain cultivars. Alkalinity decreased the solubility of micronutrients especially iron and zinc resulting in chlorosis and stunted the plant growth and development. The authors came to the conclusion that marigolds can withstand moderate

salinity, but they are very vulnerable to alkaline irrigation, and slightly acidic water  $\text{pH} \approx 5.5$  is the best to provide the marigolds with a good ornamental performance.

Mary et al. [15] introduces an IoT-based smart irrigation system to grow plants in a greenhouse that integrates the soil moisture measurements with the NDVI (Normalized Difference Vegetable Index) measurements to monitor the health of plants and to optimize the water usage. The system regulates the soil moisture with sensors and closed-loop control to keep the moisture content inside the safe range and to counter the water stress in crops of *Capsicum annum*. The experiments indicated that monitoring based on NDVI increased crop vigor, whereas the foliar glycine betaine treatment increased the drought tolerance. The paper identifies NDVI as an important plant health evaluation parameter as well as irrigation regulation. Omar et al. [16] developed a smart irrigation system based on a LoRaWAN which was proposed to implement within rose greenhouse to monitor the soil moisture, temperature, and humidity levels to study and enable automatic water supply. The system incorporates CubeCell nodes, solenoid valves and ChirpStack to handle data in the cloud and visualize it. Automated irrigation works on the principle of pausing when the humidity is lower than 60% and resuming when the humidity is greater than 75% which keeps the soil in an ideal condition. Cultivation tests were carried out in an Ecuadorian flower farm, and proved to be stable in terms of environmental control, water savings and quality of the rose. The project demonstrates that IoT was effective in attaining precision irrigation and sustainable floriculture.

### III. METHODOLOGY

The *Smart Irrigation System for Protected Flower Cultivation Using Real-Time Sensing and Automation* incorporates the logical design and interconnection of hardware, software, and cloud-based components of IoT. The system will be configured so that it will gather environmental information, do the analysis on the ESP32 microcontroller and automatically manage irrigation in regard to the real-time soil and climate conditions. The work flow in general is as represented in Fig. 1 and as discussed in the subsections below.

#### A. System Overview

The system proposed makes use of ESP32 microcontroller as the processing and communication unit. It connects with different environmental sensors to measure soil moisture, temperature, humidity, pH, and water turbidity of which they are important parameters to allow healthy flower growth in polyhouses. The data obtained is computed immediately on ESP32 and also sent to Blynk Cloud platform through Wi-Fi to display in real time and control through handset. The irrigation pump, which is linked to the relay module is automatically switched on or off by soil moisture levels.

This design provides a two-way interaction between the field sensors and the user using the Blynk mobile app. The user will be able to view real time data, see trends of the parameters and override any automated activities should there be a need

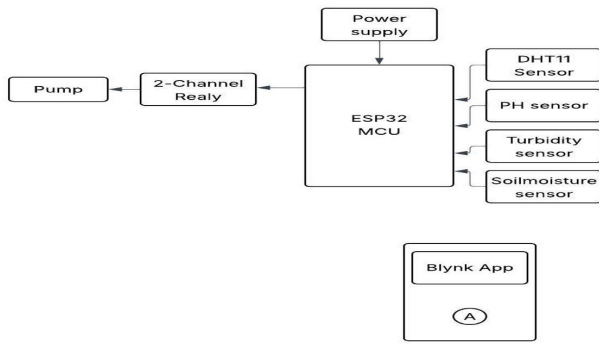


Fig. 1. System Architecture of Smart Irrigation System

thereby integrating automation and manual control to maintain reliability.

### B. Hardware Design

The ESP32 microcontroller is interconnected with several sensors in the hardware design, which is presented in Fig.1. Each component has a specific function as described below:

- 1) **ESP32 Microcontroller:** It is the control and communication unit with a dual-core processor and inbuilt Wi-Fi.
- 2) **DHT11 Sensor:** This sensor is used to measure the ambient temperature and humidity to track the changes in a micro-climate within the green house.
- 3) **Soil Moisture Sensor:** It measures the volumetric water content of soil; the ESP32 would initiate irrigation once the sensor readings were less than 40%.
- 4) **pH Sensor:** Measures the acid or alkalinity of the soil to check the best nutrient absorption by the various flower species.
- 5) **Turbidity Sensor:** Measures the level of clarity and water quality to avoid the use of dirty water to irrigate the farm.
- 6) **Relay Module:** This is a switching unit that is going to be used to automate the pump depending on ESP32 instructions.
- 7) **Water Pump:** Waters the plants in line with the signal of the relay.

The sensors are attached to specific GPIO pins (DHT11 → GPIO4, Soil Moisture → GPIO32, pH Sensor → GPIO34, DHT11 → GPIO4, Turbidity → GPIO35, Relay → GPIO18). A controlled power supply of 5V to 3.3V is used so that all components are safe. It is a small and power saving circuit and is intended to be deployed in the greenhouse on a long-term basis.

### C. Software Implementation

The system software was created utilizing the Arduino IDE since it fully supports programming for ESP32 and the required Serial communication. The programming structure allowed for automatic reconnection to Wi-Fi, automated sensor calibration, and intelligent use of the irrigation pump. The raw analog returned from each sensor underwent analog to digital

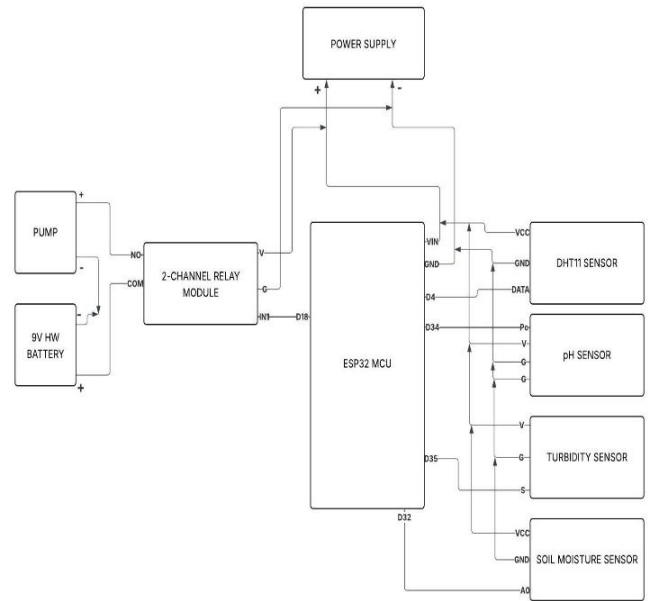


Fig. 2. Circuit Diagram showing ESP32 and Sensor Connections

conversion and conversion from raw values to meaningful physical quantities including percentage moisture, pH value, and turbidity level. This facilitated accurate monitoring and decision making for the greenhouse environment.

Temp Status  
N/A  
Humidity Status  
N/A  
Soil Status  
Dry  
pH Quality  
Poor  
Turbidity Quality  
Poor

Fig. 3. Real-time Monitoring Dashboard with Status

The control algorithm runs the following logic:

- 1) The ESP32 continuously reads sensor data at predefined time intervals.
- 2) If the soil moisture value  $\leq 40\%$ , the relay is triggered, switching the Pump ON.
- 3) If the soil moisture value  $> 40\%$ , the Pump OFF condition is executed.
- 4) At the same time, all data is sent to the **Blynk Cloud** using virtual pins (V0-V10)

- 5) The **Blynk App** dashboard provides live readings and relay status, as well as the environmental status as *Good*, *Moderate*, or *Poor*.

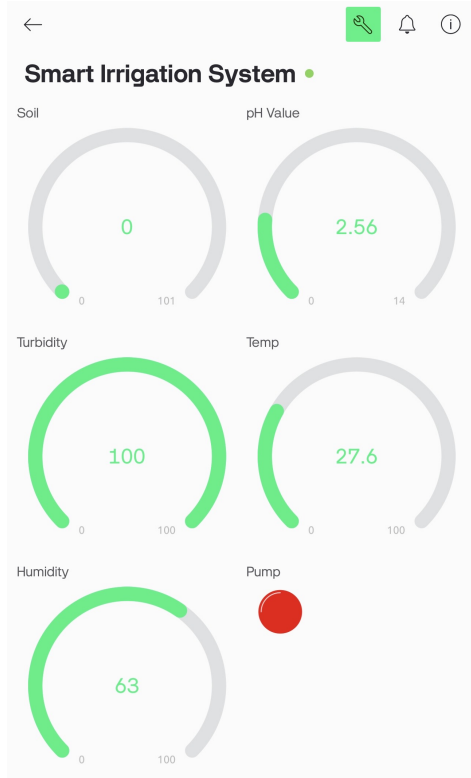


Fig. 4. UI of Real-Time Monitoring and Control

All sensor readings and the status of the system are sent to Blynk Cloud through virtual pins (V0-V10). Blynk mobile application displays the most recent values, pump status, and color-coded quality indicators (Good, Moderate, Poor). Users can remotely activate/deactivate the pump manually via the app interface, which helps with ease and reliability of operation.

#### D. Data Flow and Communication

The procedure of communication follows a hierarchical flow of data, which starts with the acquisition of sensor data, processing the data on the ESP32, and ending with the communication of data to the Blynk IoT platform. Figure 4 (Data Flow Diagram) illustrates the described flow.

The communication protocol of the ESP32 uses Wi-Fi over TCP/IP to achieve reliability of data transmission. Each reading is time-stamped and logged on the Blynk cloud for visualization and analysis. This is important for data driven decisions any time during the performance of the equipment and for long application cycles in cultivation.

#### E. Experimental Setup and Working

The prototype was tested in a small protected environment to simulate greenhouse conditions. The sensors were incorporated into the soil bed adjacent to flower plants and connected

to a regulated DC supply. The ESP32 was configured to interface with the Blynk cloud via a local Wi-Fi network. When the soil dryness condition was met, the system automatically turned on the pump and provided real-time feedback on the app regarding the pump status.

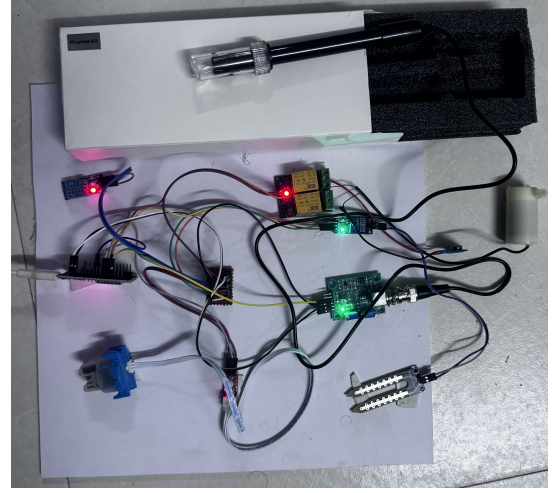


Fig. 5. Smart Irrigation System

This real-time control preserved water and provided adequate moisture throughout the area. The data obtained from the Blynk cloud demonstrated that all parameters were stable and accurate, indicating reliability and efficiency of the system.

## IV. RESULTS AND DISCUSSION

The proposed *Smart Irrigation System for Protected Flower Cultivation Using Real-Time Sensing and Automation* was designed, implemented, and tested under the conditions of controlled environment. The primary aim of the system was to have automated and real time irrigation with optimal soil and environmental conditions within the greenhouse. The outcomes of the conducted experiments prove that the system can optimize water consumption, balance soil moisture, and constantly monitor the state of affairs using the Blynk IoT platform. The results are elaborated below.

#### A. Sensor Data Acquisition and Monitoring

The sensors were all connected to the ESP32 microcontroller and were able to relay the information to the Blynk Cloud via Wi-Fi. The water level sensor was very effective in detecting changing water levels, and the sensor indicated the percentages of the moisture level in the soil. The ESP32 was used to activate the relay module when the humidity was below 40 percent of the threshold and then the water pump was turned on. The pump went off automatically when the threshold was reached.

At the same time, the sensor of DHT11 was able to read necessary temperatures and humidity and the pH sensor ensured soil acidity of between 6.0 and 7.0, which are very favorable to flower crops. The turbidity sensor was used to determine the purity of the irrigation water through the water

clarity. All the parameters were displayed in real-time by the Blynk dashboard with the color indicators of Good, Moderate, or Poor condition.

### B. Automated Irrigation and Control Logic Validation

The embedded control logic on ESP32 was tested on varied soil moisture conditions. The relay module was activated immediately the moisture of the soil fell below 40 percent to turn on the water pump. The pump was off once the required amount of moisture had been attained. This test proved that the system was responsive and reliable. The Wi-Fi connection was also stable and the auto-reconnection feature also ensured stable operation.

### C. System Reliability and User Interaction

The Blynk mobile dashboard was a trusted and user-friendly interface to farmers. The user was able to see real time data, examine previous trends, and control the pumps by hand when necessary. The hybrid design, which integrates automation and manual control, will provide flexibility to the circumstances of an emergency or crop-specific situations. The modular sensor system has a possibility to add light intensity or CO<sub>2</sub> sensors or nutrient sensors later to enable more sophisticated automation.

### D. Effect of Soil pH on Smart Irrigation System Performance

To examine the effects of the Smart Irrigation System on soil moisture retention, plant health, and efficiency of irrigation system, the performance of the Smart Irrigation System in various soil pH was analyzed and determined. pH is a critical factor in determining nutrient availability, water uptake, and physiological reaction of flowers growing under controlled conditions.

TABLE I  
VARIATION OF SYSTEM PARAMETERS, PLANT CONDITION, AND  
IRRIGATION RESPONSE UNDER DIFFERENT PH LEVELS

| pH Level | Soil Moisture (%) | Temperature (°C) | Humidity (%) | Turbidity (NTU) | Plant Condition | Irrigation System Response                   | Observation Summary                                                |
|----------|-------------------|------------------|--------------|-----------------|-----------------|----------------------------------------------|--------------------------------------------------------------------|
| 4.0      | 30                | 25               | 55           | 25              | Poor            | Pump ON frequently (soil retains less water) | Acidic soil reduces nutrient uptake; plant growth stunted.         |
| 5.0      | 38                | 26               | 58           | 20              | Moderate        | Pump ON intermittently                       | Improved absorption but mild stress symptoms observed.             |
| 6.0      | 45                | 27               | 60           | 15              | Excellent       | Pump cycles balanced                         | Ideal nutrient absorption and soil moisture; best growth observed. |
| 7.0      | 42                | 28               | 62           | 18              | Good            | Moderate irrigation                          | Stable growth, slight decline in flower color intensity.           |
| 8.0      | 35                | 29               | 64           | 28              | Poor            | Pump ON frequently due to nutrient imbalance | High alkalinity causes chlorosis and reduced bloom quality.        |

In the table I, the system has optimum performance at a pH of 6.0-7.0 whereby the soil nutrients and moisture are balanced leading to active plant growth and low stress measures. The ESP32 powered relay is a very effective way of keeping the irrigation cycles under check to avoid under- and over-watering. When pH 4.0 and 5.0, the acidic environment results to low nutrient absorption, which results in high frequency of pump action because of the inability of the soil water to retain nutrients and the observation of stress in plant foliage. On the other hand, alkalinity causes nutrient lock out at pH 8.0, particularly in the essential micronutrients such as iron and zinc, leading to chlorosis, reduced flower coloration and poor soil moisture stability.

In general, the findings indicate that the steady pH level of soils within the slightly acidic to neutral range does not only

promote the vitality of plants but it also increases the accuracy of irrigation and water-use efficiency. The combination of pH monitoring and the IoT-based control logic is used to make the system reshape the irrigation scheduling in real-time, according to the environmental and crop-specific requirements, which will, in the end, enable sustainable precision agriculture.

### E. Summary of Experimental Observations

Table II summarizes the key parameters observed during testing, along with their measured ranges, optimal thresholds, and system responses.

TABLE II  
EXPERIMENTAL READINGS AND SYSTEM RESPONSES

| Parameter             | Measured Range | Threshold | Action Taken                    |
|-----------------------|----------------|-----------|---------------------------------|
| Soil Moisture (%)     | 25-75          | 40%       | Pump ON $\leq$ 40%, OFF $>$ 40% |
| Temperature (°C)      | 24-31          | 20-30     | Monitored only                  |
| Humidity (%)          | 45-70          | 50-65     | Monitored only                  |
| pH Level              | 6.0-7.2        | 6.5-7.0   | Monitored for nutrient health   |
| Water Turbidity (NTU) | 5-50           | $<$ 20    | Alert if $>$ 30 NTU             |

The findings affirm the assumption that the IoT-based smart irrigation system offers real-time, precise, and automated control of the water consumption that can be used to support sustainable and precision farming. Through a combination of several sensors and a visualization system that harbors cloud computing, the system lowers the level of human error, decreases and maximizes the use of water, and improves the quality of crops. The modularity and scalability of the hardware can be upgraded over time, with possibilities of future upgrades to include machine learning-driven prediction, solar integration, and AI-driven crop management. In this way, the system itself is a viable and efficient approach to the current trend of greenhouse and floriculture.

## V. CONCLUSION AND FUTURE SCOPE

### A. Conclusion

The SSmart Irrigation System for Protected Flower Cultivation Using Real-Time Sensing and Automation is a good example of how the IoT technology, sensor networks, and automation be implemented to efficiently control water in agriculture. The system can constantly measure the key environmental conditions by using an ESP32 microcontroller as the central processing and communications device, such as soil moisture, temperature, humidity, pH, and water turbidity. The functionality of visualizing real-time data, remote monitoring, and manual control of the system via a smartphone interface is introduced through the addition of Blynk IoT platform.

The system was tested by deploying experimentally and with ideal soil moisture, it only switches on the irrigation pump when the moisture goes past the optimum threshold, hence, saving water, and enhancing efficiency. optimum threshold, hence, saving water, and enhancing efficiency. Further, there were also the findings that showed an improvement in crop quality, consistency, and a decrease in manual labor. Moreover, monitoring and automation Cloud computing can

be used to achieve scalability, as the system can be reconfigured to serve with minimal effort in large-scale agricultural applications, including commercial greenhouses and smart farms. Finally, it is concluded that the present investigation offered a cost-effective, dependable, and greener approach in implementing an environmental sustainability and precision farming to the contemporary farming operations.

### B. Future Scope

As much as the current system has installed automated operations to skin-deep level and control remotely, it may take enormous steps in intelligence, scaling and maintenance. Future development can be the addition of weather forecasting APIs to dynamically alter irrigation schedules by forecasting models, like rainfall predictions, humidity variations, temperature patterns, etc. Moreover, this can be complemented by solar power, which will render the whole system self-sufficient to the agricultural sites, rural and otherwise, which themselves are also completely off-the-grid.

Machine learning and artificial intelligence can be also incorporated in the system as predictive irrigation or data-driven decision-making. As an example, machine learning models can be trained on previous soil and weather data to ensure that the system could automatically detect the crop-specific patterns and automatically run the irrigation cycles with the highest yield and minimum waste. Also, user-friendly pasture operations can have voice-assistive or AI-based interfaces, where users can easily control the devices by voice and have the interface in their native language. Finally, the system may be extended by integrating with low-power wireless technology like LoRaWAN or Zigbee that may enable the user to control multiple zones or polyhouses simultaneously.

In the long run, the proposed system can evolve into a complete autonomous farm control system that will allow controlling not only irrigation, but also nutrient, pest status, and energy levels, among others. This development into a single irrigation controller rather than a fully integrated and connected smart-farming system would be significant towards precision agriculture, sustainability, and global water conservation program.

### REFERENCES

- [1] A. N. Hussain and S. Saradha, "An iot sensor network system- based precision irrigation system deploying with collaborative sensor techniques for the optimumlow - cost yield," in 2022 International Conference on Augmented Intelligence and Sustainable Systems (ICAISS), pp. 1048–1053, 2022.
- [2] A. G. Mohana, S. S. K. A. S. B S, and N. K, "Efficient smart irrigation system using internet of things (iot) for agriculture crops," in 2025 International Conference on Electronics and Renewable Systems (ICEARS), pp. 680–685, 2025.
- [3] A. N. P J, M. Sreejith, K. R. S. Tiwari, A. P. A, J. K R, and S. Nalinakshan, "Empowering agriculture with real-time iot monitoring," in 2025 Emerging Technologies for Intelligent Systems (ETIS), pp. 1–5, 2025.
- [4] B. Sowmya, J. Jain, T. Kailas, and G. Rahul Malik, "Integrating iot and hybrid energy for sustainable smart irrigation systems in agriculture," pp. 28–35, 08 2025.
- [5] D. Abhiram, K. D. Reddy, P. K. Reddy, and M. Belwal, "Iot-based environmental monitoring system," in 2025 3rd International Conference

- on Intelligent Data Communication Technologies and Internet of Things (IDCIoT), pp. 764–769, 2025.
- [6] A. N and R. M, "Precision agriculture: Improving water stress forecasting in tomato plants using bioristor data and advanced ml techniques," in 2024 First International Conference for Women in Computing (In-CoWoCo), pp. 1–6, 2024.
- [7] N. Shuhaimi, C. C. Chew, T. Gaik, and K. Tee, "Smart irrigation monitoring and controlling system with the internet of things (iot)," pp. 43–47, 08 2024.
- [8] I. Rajapaksha, T. Gamage, I. Eranga, A. Dharmasena, and K. Gamage, "Smart irrigation system for domestic homes," in 2022 4th International Conference on Smart Sensors and Application (ICSSA), pp. 96–101, 2022.
- [9] S. Chadli, B. Bengherbia, K. Said, Y. Remram, N. Oudahi, and A. Toubal, "Smart irrigation system with low-power wsns for sustainable water management in agriculture," pp. 1–6, 05 2025.
- [10] T. Mallikarathne, P. M. Perera, G. G.A.H.K.M, G. H.T., U. W.V.A, and M. S.A.D.H.P, "Wireless sensor network based smart irrigation system using iot and data fusion," in 2024 4th International Conference on Advanced Research in Computing (ICARC), pp. 259–263, 2024.
- [11] V. LR, T. P, R. S, and U. K K, "Assessment of soil and water quality status of rose growing areas of rajasthan and uttar pradesh, india," Journal of Horticultural Sciences, vol. 16, p. 280–286, Dec. 2021.
- [12] M. Markovi´c, V. Gali´c, V. Težak, M. Ravli´c, B. Zeljko, I. Jug, and L. Gali´c, "The irrigation water ph has a dominant impact on the growth and stress markers of bigleaf hydrangea," Applied Sciences, vol. 15, p. 8773, 08 2025.
- [13] R. Gentili, R. Ambrosini, C. Montagnani, S. Caronni, and S. Citterio, "Effect of soil ph on the growth, reproductive investment and pollen allergenicity of ambrosia artemisiifolia l.," Frontiers in Plant Science, vol. 9, p. 1335, 09 2018.
- [14] L. Valdez-Aguilar, C. Grieve, and J. Poss, "Salinity and alkaline ph in irrigation water affect marigold plants: I. growth and shoot dry weight partitioning," HortScience: a publication of the American Society for Horticultural Science, vol. 44, 10 2009.
- [15] M. Cervera-Díaz, M. F. León-Chávez, C. Santiesteban-Toca, and C. Lozoya, "Greenhouse Smart Irrigation Based on Soil Moisture and Vegetation Index Measurements," in 2023 IEEE Conference on Technologies for Sustainability (SusTech), pp. 20–24, 2023.
- [16] O. Moreira, G. Rocano, E. Ordoñez, J. Inga-Ortega, D. Rivas-Lalaleo, R. Clotet, K. Belloso, J.-I. Castillo-Velazquez, and M. Huerta, "Development of an Intelligent Irrigation System for a Rose Cultivation Based on IoT," in 2024 IEEE ANDESCON, pp. 1–6, 2024.